

HRV and Training Readiness

Reference notes on heart rate variability (HRV) as a recovery signal. Paraphrased from sports-science reviews and wearable-industry education material. Not medical advice.

What HRV represents

Heart rate variability is the natural variation in time between consecutive heartbeats, reflecting the balance between the body's stress-response (sympathetic) and rest-and-digest (parasympathetic) nervous system activity. A higher HRV, relative to a person's own baseline, generally indicates the body is well-recovered and ready to handle training stress; a lower HRV relative to baseline suggests accumulated fatigue, stress, illness, or insufficient recovery.

Why HRV should always be compared to a personal baseline

Absolute HRV values vary enormously between individuals due to age, fitness level, and physiology, so a single reading means little on its own. What matters for a daily readiness assessment is the trend relative to that person's own rolling average (for example, the last 7-14 days), not a universal cutoff.

A single low reading is not very meaningful by itself, since normal day-to-day fluctuation happens even in well-recovered people. A multi-day downward trend is a much stronger signal than any one data point.

Typical interpretation ranges (relative to personal baseline)

- At or above baseline: normal readiness, training as planned is generally reasonable.
- Slightly below baseline (roughly within one typical day-to-day fluctuation): mild fatigue, consider a lighter or shorter session.
- Notably below baseline for one or more days, especially alongside poor sleep or elevated resting heart rate: reduce intensity, favor active recovery or rest.
- Persistently suppressed for many days in a row: a sign to lean toward extended rest and, if it continues, mention checking in with a healthcare provider or coach.

Related signals that add context

Resting heart rate, elevated well above a person's baseline, often accompanies low HRV and adds confidence to a 'low readiness' assessment. HRV suppressed for 24-72 hours after a hard session is a normal, expected part of recovery, not automatically a red flag on its own.

Guidance for the assistant's scoring logic

When combining HRV with sleep and training load into a recovery score, HRV trend relative to baseline should be treated as one input among several rather than the sole determinant, since it can be affected by non-training factors like stress, alcohol, illness, or poor sleep the same night.